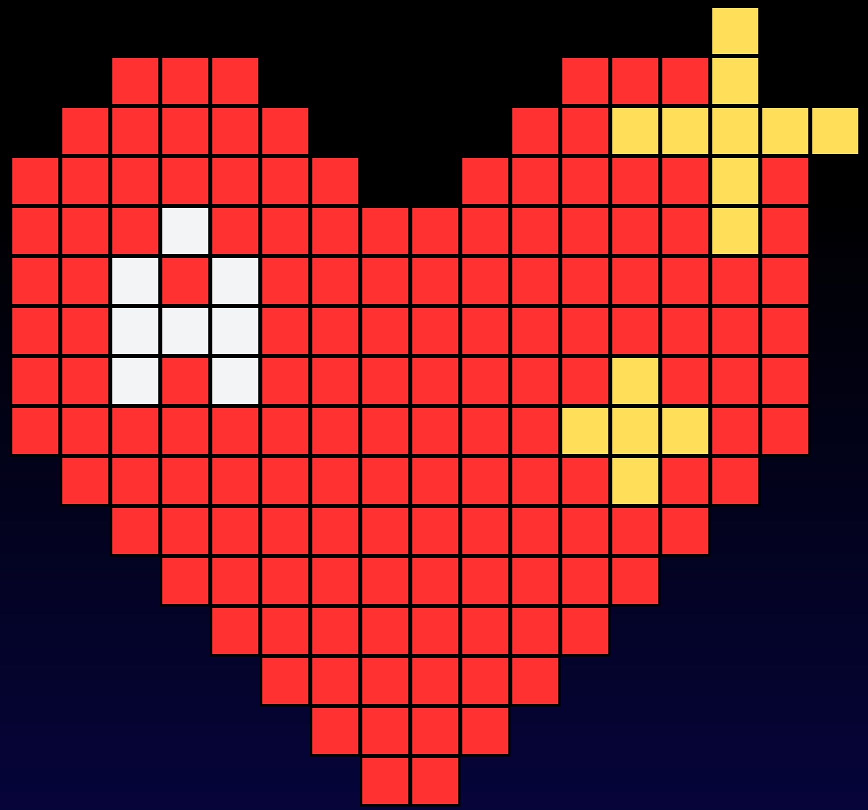
F.O.P

MINI-PROJECT

// Aroga
// MODERN FITNESS TRACKER
// USING C

// Faculty: Dr. Uma Pujeri



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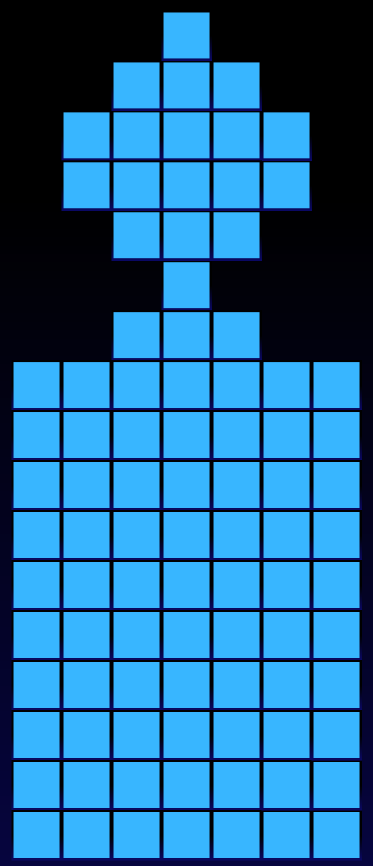


INTRODUCTION

// In modern society, maintaining physical fitness has become increasingly important due to sedentary lifestyles and unhealthy eating habits. Many individuals fail to track their calorie intake and physical activity, leading to obesity and lifestyle-related diseases.

A fitness tracking system helps individuals :-

1. Monitor their Body Mass Index (BMI)
2. Track daily calorie intake
3. Calculate calories burned through exercises
4. Maintain a calorie balance
5. Predict weight gain or loss trends

This project provides a simple yet effective solution using C programming concepts to assist users in managing their fitness journey.

OBJECTIVES

Develop a fitness tracking application that allows users to manually input and monitor health and activity data.

Includes:

1. Manual entry of workouts, steps, calories, weight, Height, age.
2. Demonstration of Data storage using File Handling of C.
3. Stores data with daily records in text files with date and time.
4. Uses Date-Time Module to entry the data of everyday.
5. Uses text files to as source to save the data

FEATURES

User Profile Management

- Takes Height, weight, age, Gender and Activity level as input.
- Calculates the BMI using the Formula and tells whether the person is underweight, overweight, healthy weight or Obese.

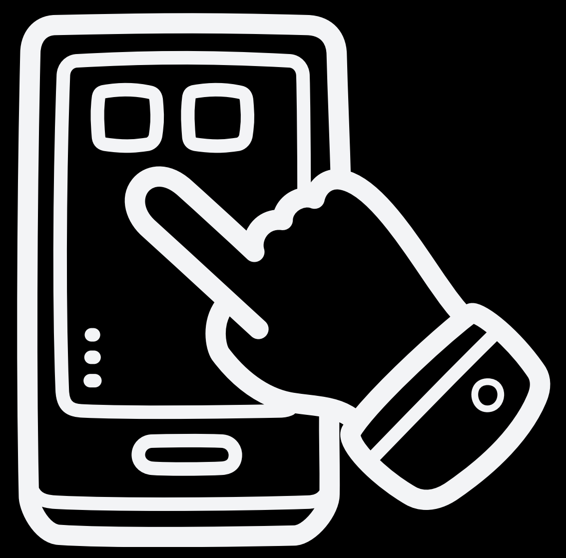
BMI = Weight / (Height * Height).

Tracks Calories Burned for :

- For now, it tracks calories burned during Aerobic and Calisthenic Exercises.
- Saves the data in a text file of amount of calories burned with exercise done on that day.
- Tells the amount of weight lost by person by burning calories.
- User can access the exercise report from menu by entering date

FEATURES

Food Calorie Tracker



- Fetches data of Calories of Food from Text based database and tells data for :
  - Break-Fast
  - Lunch
  - Dinner
- Calculates the total Calorie Intake of the user of that day.
- Calculates the Net Calorie person took and matches that with recommended calorie intake as per **Milfin St. Jeor Equation** and Activity Level of the Person.
- Generates a Daily Report of Food intake of person and Calories burned by person
- Has menu to access the report of specific date.
- Stores all the data in text file that can be accessed through code

METHODOLOGY

Step 1:
User Enters Height, Age, Weight, Gender, Activity Level which can be updated and initialized

Step 2:
Calculates BMI and stores it with UserInfo.txt

Step 3:
For Exercise Tracker, User Enters reps, sets and duration of exercise whereas for aerobic exercises like running etc. user enters distance, time
It calculates the calories burned by the user. Also the weight lost by the user

Step 4:
User enters the food items he ate during breakfast, lunch and dinner.
User can also enter water intake.
User enters the Quantity Consumed
The program fetches data from food.txt

Step 5:
Based on the Net calorie intake by the user the program tells whether a person is calorie deficit or calory surplus.

Step 6:
From menu user can access his generated report by entering date for both exercise and calorie tracker.

Step 7:
Finally saves the data into their respective files

FUNCTIONS AND MODULES USED

Date and Time module

File Handling functions of C

User Defined Data type (struct)

Arithmetic Operations

Conditional Statements like if-else and Switch Case.

ADVANTAGES

1. Easy to use

2. Encourages Healthy Lifestyle

3. Tracks Net calories by tracking gained calories and burned calories

4. User can be aware of whether he is losing or gaining weight

5. Maintains daily records just like a Diary

6. Cost effective solution

LIMITATIONS

It is not fully accurate because calorie values are approximate and they tend to change depending on food.

Does consider metabolism, but may vary person to person.

Manual Data entry required

Less Variety of foods due to limitation of using text file.

FUTURE ENHANCEMENTS

1. Adding Graphical User Interface
2. Using Graphs to track user data
3. Calculating Macros for given Meal.
4. Adding Sleep as a parameter
5. Adding Goal Selection system in which person can specify whether he wants to lose, gain or keep it's weight stable
6. Integrating the code with Online API so program will get access to all variety of foods.

CONCLUSION

The Fitness Tracker project successfully demonstrates the application of C programming concepts in solving real-world health management problems.

By integrating BMI calculation, calorie tracking, weight prediction, and file storage, the system provides a structured approach to monitoring fitness.

Although the current version is console-based and limited in scope, it has strong potential for further expansion into a complete health management system. The project effectively combines programming logic with practical health applications.

A probable all-in-one health management system...?

OUTPUT

C:\Users\ojhar\OneDrive\Desktop\C Code\FINAL\MyFitnessTracker.exe

```
===== MyFitnessTracker =====
1) User Info (Init / Edit)
2) Exercise Tracker
3) Calorie Tracker
4) Exit
Enter choice:
```

OUTPUT

C:\Users\ojhar\OneDrive\Desktop\C Code\FINAL\MyFitnessTracker.exe

```
===== MyFitnessTracker =====
1) User Info (Init / Edit)
2) Exercise Tracker
3) Calorie Tracker
4) Exit
Enter choice: 1

1) Initialize User Info
2) Edit User Info
Enter choice: 1

===== User Info Setup =====
Enter your Age: 18
Enter Your Height in cm: 181
Enter Your Weight in kg: 60.95
Enter your Gender (M/F): M

Activity Level:
1) Sedentary (little or no exercise)
2) Lightly Active (exercise 1-3 days/week)
3) Moderately Active (exercise 3-5 days/week)
4) Very Active (exercise 6-7 days/week)
5) Extremely Active (physical job or training 2x/day)
Enter your activity level (1-5): 3
Your BMI is 18.60
User info saved successfully!
```

OUTPUT

C:\Users\ojhar\OneDrive\Desktop\C Code\FINAL\MyFitnessTracker.exe

```
Enter choice: 2

1) Add Exercise Data
2) View Exercise Data by Date
Enter choice: 1

User info loaded from file!

===== Exercise Tracker =====
Your Current Weight: 60.95 kg
Enter the no. of Total Exercise you did Today :2

Exercise 1:
Enter The Body Weight Exercise You Did :
 1) Press 1 For Pushups
 2) Press 2 For Squats
 3) Press 3 For Pull-ups
 4) Press 4 for Crunches
 5) Press 5 for Sit-Ups
 6) Press 6 for Running
 7) Press 7 for Walking
 8) Press 8 for Swimming
 9) Press 9 for Cycling
 Your Choice :
```

OUTPUT

C:\Users\ojhar\OneDrive\Desktop\C Code\FINAL\MyFitnessTracker.exe

```
User info saved successfully!

===== MyFitnessTracker =====
1) User Info (Init / Edit)
2) Exercise Tracker
3) Calorie Tracker
4) Exit
Enter choice: 3

== Calorie Tracker ==
1) Add Meal
2) View Today's Report
3) View Report By Date
4) Set Water Intake
Enter choice:
```

OUTPUT

C:\Users\ojhar\OneDrive\Desktop\C Code\FINAL\MyFitnessTracker.exe

```
== Calorie Tracker ==
1) Add Meal
2) View Today's Report
3) View Report By Date
4) Set Water Intake
Enter choice: 1

Select Meal:
1. Breakfast
2. Lunch
3. Dinner
Enter choice: 1
Enter food name: poha
Enter number of plates: 1
Added 340.00 kcal.

1. Add Another Food
2. Change Meal
3. Main Menu
Enter choice:
```

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2. Internet Browser for Getting to know more and for formulas and Equations of Project
3. Basic Calorie Burn Estimation Methods
4. Online Food Calorie API Documentation
5. C Programming Documentation
6. File Handling by Code_with_Harry on Youtube.

// Team Aroga, signing out!